

Differential Geometry Notes

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1 Preliminaries

Definition 1. A *submanifold* of M is a subset $N \subseteq M$ that is a manifold

Definition 2. The *tangent bundle* $TM \subseteq \mathbb{R}^{2k}$ of M is defined as $\{(p, v) : p \in M, v \in T_p M\}$

Claim 1. TM is a manifold of dimension $2n$

2 Notions associated with the derivative

Definition 3. Define $\text{grad} \in L(C^\infty(M), \mathcal{X}(M))$ by $\langle v, \text{grad}_f(p) \rangle = Df_p(v)$ for all $v \in T_p M$

Definition 4. Define $\text{Hess}_p \in L(C^\infty(M), T_p M^{(2)})$ by

$$f \rightarrow ((v, w) \rightarrow \langle \nabla_v X_f, w \rangle)$$

Claim 2. $\text{range Hess}_p \subseteq (T_p M)_{\text{sym}}^{(2)}$

Claim 3. The map $X, Y \rightarrow D_X Y$ is bilinear and has the following properties:

- $D_f X Y = f(D_X Y)$
- $D_X(fY) = f(D_X Y) + (D_X f)Y$
- $D_X Y - D_Y X = [X, Y]$
- $D_X \langle Y, Z \rangle = \langle D_X Y, Z \rangle + \langle Y, D_X Z \rangle$

Claim 4. Suppose $\phi : \Omega \rightarrow V$ is a local parametrization. Then $D_{\phi_i} \phi_j = \frac{\partial \phi}{\partial x_i \partial x_j}$

3 Pullback by a diffeomorphism

Suppose $F : M \rightarrow N$ is a diffeomorphism

Claim 5. Suppose $v \in T_p M$. Then for all $f \in C^\infty(M)$ we have $D_v(f \circ F) = (D_{F_* v})f$

Claim 6. We have $F_*[X, Y] = [F_* X, F_* Y]$

Claim 7. We have $\langle F_* X, F_* Y \rangle = \langle X, Y \rangle$

4 Riemannian metric

Definition 5. A diffeomorphism $F : M \rightarrow N$ is called a **Riemannian isometry** if at each point the derivative DF_p is an isometry from $T_p M$ to $T_p N$

Definition 6. A local diffeomorphism $F : M \rightarrow N$ at p is called a **local Riemannian isometry** at p if there is a neighborhood U such that $F(U)$ is open and $F|_U : U \rightarrow F(U)$ is a Riemannian isometry

Definition 7. Suppose $\gamma : [a, b] \rightarrow M$ is a piecewise smooth curve. A **reparametrization** of γ is given by $\gamma \circ \phi$, where $\phi : [c, d] \rightarrow [a, b]$ is a homeomorphism with a partition $c_0, c_1, \dots, c_k$ such that $\phi|_{[c_{i-1}, c_i]}$ is a diffeomorphism onto its image

Claim 8. For any γ piecewise regular, the map

$$c \rightarrow \text{length}(\gamma|_{[a, c]})$$

is a valid reparametrization, called the arc length parametrization

Suppose M is connected

Claim 9. There is a piecewise regular curve between any two points

Definition 8. Define the **Riemannian metric** by

$$d_M(p, q) = \inf\{L(\gamma) : \gamma \text{ is a piecewise smooth curve from } p \text{ to } q\}$$

Claim 10. The topology induced by the Riemannian metric is the same as the subspace topology M inherits from $\mathbb{R}^k$

Definition 9. A piecewise smooth curve γ from p to q is **length minimizing** if it has length $d_M(p, q)$

Claim 11. If there is a length minimizing curve from p to q then there is such a length minimizing piecewise regular curve as well

Claim 12. If M is closed in $\mathbb{R}^k$ then there is a length minimizing curve between any two points

Claim 13. Every Riemannian isometry is an isometry with respect to the Riemannian metric

Claim 14. Suppose F_1, F_2 are local Riemannian isometries with $F_1(p) = F_2(p)$ and $(DF_1)_p = (DF_2)_p$. Then $F_1 = F_2$

5 First fundamental form

Definition 10. The **first fundamental form** of M at p is defined by restricting the standard inner product on $\mathbb{R}^k$ to the tangent space. We have $g_p : T_p M \times T_p M \rightarrow \mathbb{R}$

Claim 15. Two manifolds M and N admit a local Riemannian isometry from p to q if and only if there are local parametrizations $\phi : \Omega \rightarrow U$ and $\varphi : \Omega \rightarrow V$ such that $g_{\phi(a)}(v_i, v_j) = h_{\varphi(a)}(w_i, w_j)$ for all $a \in \Omega$, where $v_i = \phi_*(\phi(a))$ and $w_i = \varphi_*(\varphi(a))$

6 Levi-Civita connection

Definition 11. We have $\pi_p \in \text{End}(\mathbb{R}^k)$ which is the orthogonal projection onto $T_p M$. We define $\hat{\pi}_p$ as $I - \pi_p$

Definition 12. Suppose $v \in T_p M$. Define $\nabla_v \in L(\mathcal{X}(M), T_p M)$ by $X \rightarrow \pi_p(DX)_p v$

Definition 13. Suppose $X, Y \in \mathcal{X}(M)$ are smooth vector fields. Then we have the smooth vector field $\nabla_X Y = (p \rightarrow \pi_p(D_X Y)(p))$ called the **Levi-Civita connection** of X and Y

Claim 16. The map $X, Y \rightarrow \nabla_X Y$ is bilinear and has the following properties:

- $\nabla_X Y - \nabla_Y X = [X, Y]$
- $\nabla_{fX} Y = f(\nabla_X Y)$
- $\nabla_X(fY) = f(\nabla_X Y) + (D_X f)Y$

Definition 14. Suppose $\phi : \Omega \rightarrow V$ is a local parametrization. The **Christoffel symbols** $\Gamma_{ij}^k \in C^\infty(\Omega)$ are defined by $(\nabla_{\phi_i} \phi_j)(\phi(a)) = \sum_k \Gamma_{ij}^k(a) \phi_k(a)$

Definition 15. The **curvature tensor** of X, Y, Z is defined by $R(X, Y, Z) = \nabla_X(\nabla_Y Z) - \nabla_Y(\nabla_X Z) - \nabla_{[X, Y]} Z$

Suppose $F : M \rightarrow N$ is a Riemannian isometry

Claim 17. We have $\nabla_{F_* X}(F_* Y) = F_*(\nabla_X Y)$

Claim 18. We have $\nabla_{F_* v} \circ F_* = DF_p \circ \nabla_v$

Claim 19. We have $F_*[R(X, Y, Z)] = R(F_* X, F_* Y, F_* Z)$

7 Vector fields along a curve

Definition 16. Suppose $\gamma : I \rightarrow M$ is a smooth curve. A smooth vector field along γ is a map $V : I \rightarrow \mathbb{R}^k$ such that $V(t) \in T_{\gamma(t)} M$

The vector space of smooth vector fields along γ is $\mathcal{X}(\gamma)$

Definition 17. We have $\text{cov} \in L(\mathcal{X}(\gamma))$ given by $V \rightarrow (t \rightarrow \pi_{\gamma(t)} \dot{V}(t))$

Definition 18. A vector field V along γ is called **parallel** to γ if $V \in \ker(\text{cov}_\gamma)$

Claim 20. Suppose $t_0 \in I$, $p_0 = \gamma(t_0)$ and $v_0 \in T_{p_0}M$. There is a unique parallel vector field V satisfying $V(t_0) = v_0$

Definition 19. Let $t_1, t_2 \in I$. We have the **parallel transport** map $P_{t_1 t_2} \in L(T_{\gamma(t_1)}M, T_{\gamma(t_2)}M)$

Claim 21. $P_{t_1 t_2}$ is an isometry

Claim 22. $P_{t_1 t_2}$ is invertible with inverse $P_{t_2 t_1}$

Claim 23. Suppose t_0 is in the interior of I . Then we have

$$\nabla_v X = \lim_{t \rightarrow t_0} \frac{P_{t t_0}(X(\gamma(t))) - X_p}{t - t_0}$$

8 Geodesics

Definition 20. A smooth curve γ is a **geodesic** if γ' is parallel to γ

Suppose $p \in M$ and $v \in T_p M$. Consider geodesics γ such that $\gamma(0) = p$ and $\gamma'(0) = v$

Claim 24. There is such a geodesic γ on $(-\epsilon, \epsilon)$

Claim 25. Any two such geodesics agree on their common domain

Claim 26. There is a unique **maximal geodesic** γ_v of this kind on an open interval I_v

Claim 27. $I_{cv} = \frac{I_v}{c}$ and $\gamma_{cv}(t) = \gamma_v(ct)$

8.1 Relation with length minimizing cuves

Claim 28. If γ is a geodesic then for each t we have that $\gamma|_{[a, b]}$ is length minimizing where $t \in [a, b]$

Claim 29. Suppose γ is piecewise regular and length minimizing. Then γ is smooth

Claim 30. Suppose γ is regular and length minimizing. Then the arc length parametrization of γ is a geodesic

9 Exponential map

Definition 21. We define $E_p = \{v \in T_p M : 1 \in I_v\}$

Claim 31. E_p is star convex with respect to 0

Claim 32. E_p is open in $T_p M$

Claim 33. $I_v = \{t : tv \in E_p\}$

Definition 22. We define $\exp_p \in C^\infty(E_p, M)$ by $v \rightarrow \gamma_v(1)$

Claim 34. $(D\exp_p)_0 = I$

Claim 35. $\exp_p$ is a local diffeomorphism at 0

Claim 36. $\gamma_v(t) = \exp(tv)$

Definition 23. We define $E \subseteq TM$ as $\{(p, v) : v \in E_p\}$

Claim 37. E is open in TM

Definition 24. We define $\exp \in C^\infty(E, M)$ by $(p, v) \rightarrow \exp_p(v)$

10 Normal coordinate systems

Definition 25. A neighborhood $U \subseteq E_p$ of the origin is called **prenormal** if $\exp : U \rightarrow \exp(U)$ is a diffeomorphism.

Definition 26. A **normal neighborhood** is the image under the exponential map of a star convex prenormal neighborhood

11 Injectivity radius

Definition 27. The **injectivity radius** is defined as

$$inj_p = \sup \{r > 0 : B_r \text{ is prenormal}\}$$

Claim 38. If $\rho = inj_p$, then B_ρ is prenormal

12 One parameter family of curves

Definition 28. We call $\Gamma \in C^\infty(I \times J, M)$ a **one parameter family of curves** on J

Definition 29. The **variation field** $V \in \mathcal{X}(\Gamma_0)$ of Γ is defined by $t \rightarrow \partial_1 \Gamma(0, t)$

Claim 39. We have $[cov(\partial_1 \Gamma)_s](t) = [cov(\partial_2 \Gamma)_t](s)$

Suppose $\gamma : [a, b] \rightarrow M$ denotes a regular curve from p to q

Definition 30. A **proper variation** of γ is a map $\Gamma \in C^\infty((-\epsilon, \epsilon) \times K, M)$ such that $\Gamma_0 = \gamma$ and Γ_s is a regular curve from p to q

Claim 40. Suppose $V \in \mathcal{X}(\gamma)$ with $V(a) = V(b) = 0$. Then there is a proper variation Γ of γ whose variation field is V

13 Second fundamental form

Definition 31. $N_p M$ is the orthogonal complement of $T_p M$ in $\mathbb{R}^k$

Definition 32. The **second fundamental form** of M at p is the bilinear map $S_p : T_p M \times T_p M \rightarrow \mathbb{R}^k$ given by $v, w \rightarrow [(D\pi)_p v]w$

Claim 41. $\text{range } S_p \subseteq N_p M$ for all p

Claim 42. The second fundamental form is symmetric

Claim 43. We have $S_p(X_p, Y_p) = \hat{\pi}_p((D_X Y)(p))$